

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - n78 (3.3-3.8 GHz 5G NR) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting n78 (3.3-3.8 GHz 5G NR) for UWB (Ultra-Wideband) Filter. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 2197.3 MHz and bandwidth of 258.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: n78 (3.3-3.8 GHz 5G NR)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: 36° YX LiTaO3
- Package: SiP (System in Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting n78 (3.3-3.8 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2197.3 MHz
Bandwidth (3 dB):	258.6 MHz
Insertion Loss Target:	<= 1.4 dB
Return Loss Target:	>= 12.0 dB
Out-of-Band Rejection:	>= 51.0 dB
Isolation (Duplexer):	>= 16.0 dB
Q Factor:	2145
Temperature Coefficient:	-23.3 ppm/°C
Die Size:	0.9 x 2.3 mm
Wafer Size:	6-inch
Number of Resonators:	6
Electrode Material:	W

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 2197.3 MHz, BW 258.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6592 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.4 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 388 MHz offset >= 51.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 6.4 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2025-03-12
Devices Tested: 99
Insertion Loss (meas): 1.07 dB
Return Loss (meas): 13.5 dB
Rejection (meas): 53.7 dB
Isolation (meas): 20.9 dB
Q Factor (meas): 2145
Group Delay Variation: 3.7 ns
Power Handling: +35 dBm
Die Yield: 91.0%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated excellent performance for n78 (3.3-3.8 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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